Jinchen Zhang

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EDUCATION

Tsinghua University **Beijing, China**
B.S. in Automation 09/2023 - 06/2027 (expected)
Relevant Courses: Calculus, Linear Algebra, Random Mathematics and Statistics, Discrete Mathematics, Operations Research, Computer Languages and Programming, Data Structures, Object-Oriented Programming Training, Computer Networks and Applications, Pattern Recognition and Machine Learning, Intelligent Speech Processing, AI for Science, Intelligent Measurement System

PUBLICATIONS AND PREPRINTS

[1] *Strength Controls Change How Interventions Compare Across Sparse Autoencoder Codes.*
Jinchen Zhang*, Zheyuan Liu*, Yihan Zhu, and Meng Jiang. 2026.
arXiv:2610.26489. Under review at ACL Rolling Review (ARR). **Equal contribution.*

[2] *Training Negatives Still Matter After Calibrating Answerability Readouts.*
Jinchen Zhang, Zheyuan Liu, Yihan Zhu, and Meng Jiang. 2026.
arXiv:2610.26461. Under review at ARR.

RESEARCH EXPERIENCE

Trustworthy LLMs: Interpretability and Answerability **Notre Dame, IN**
Advised by Prof. Meng Jiang, DM2 Lab, University of Notre Dame 07/2026 - Present

- Led experiments and manuscript writing for two first/co-first-author papers [1, 2].
- **SAE interpretability [1]:** Compared interventions across five Gemma/Llama experimental setups, showing that intervention-strength controls can change comparisons across SAE codes.
- **Answerability estimation [2]:** Analyzed predictors of whether a passage supports an answer using Qwen and Gemma, showing that training negatives affect predictions even after calibration.

Visual Instruction Pretraining (ViTP) for Human Sensing **Remote**
Advised by Prof. Yuchen Liu, North Carolina State University 01/2026 - 02/2026

- Constructed an MMFi training dataset of 1,946 examples linking depth frames, action descriptions, and skeleton cues; applied segment sampling, sequence-level train/validation splits, and corrupted-frame filtering.
- Adapted ViTP/InternVL2.5 models to investigate language-guided visual learning for action recognition; fine-tuned vision/projection layers with a frozen LLM and developed multi-frame evaluation.

Efficient Sparse Modeling for Vision-Language Models **Remote**
Advised by Prof. Yinglun Zhu, University of California, Riverside 07/2025 - 10/2025

- Rebuilt reproducible visual-token compression pipelines integrating graph routing, token merging, SSMs, and token-aligned context reinjection on single-GPU HPC.
- Designed matched controls across 45 routing and 61 selection-intervention configurations; extended CLIP classification experiments to Qwen2-VL-2B on POPE, generating 9,600 responses.
- Measured end-to-end latency and compared quality using image-grouped paired bootstrap; found only 0-1.4% speedup from post-encoding compression and no semantic-GCN advantage.

Deep Learning-Based Vehicle Trajectory Prediction **Beijing, China**
Advised by Prof. Li Li, Tsinghua University 09/2024 - 09/2025

- Developed a deep learning framework for vehicle trajectory prediction to improve prediction accuracy and real-time performance in dynamic traffic environments.
- Built a visualization system that dynamically displays predicted trajectories alongside road layouts and vehicle states, enhancing the interpretability of model outputs.
- Utilized BEVFormer to assist in a comparative experiment; standardized experimental configurations and automated logging pipelines to support consistent comparisons across runs.

SKILLS

Programming: Python, C/C++
Research Skills: Experimental design, data analysis, model evaluation, academic writing
Languages: Chinese (native); English (proficient); Spanish (basic communication)